SciLifeLab

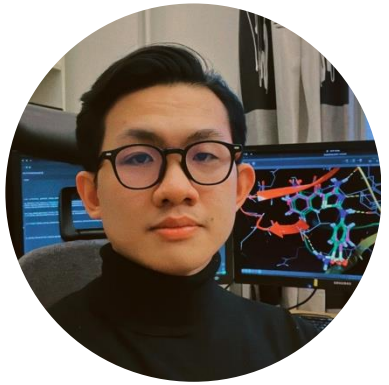
pharmb.io

AI Agents in Life Sciences

Practical Tutorials into AI Agents







shorturl.at/0GpF5

Scan to get the workshop code link

AI Agent in Life Science with LangGraph

”Hello World” – An introduction and tutorial



Presenter:

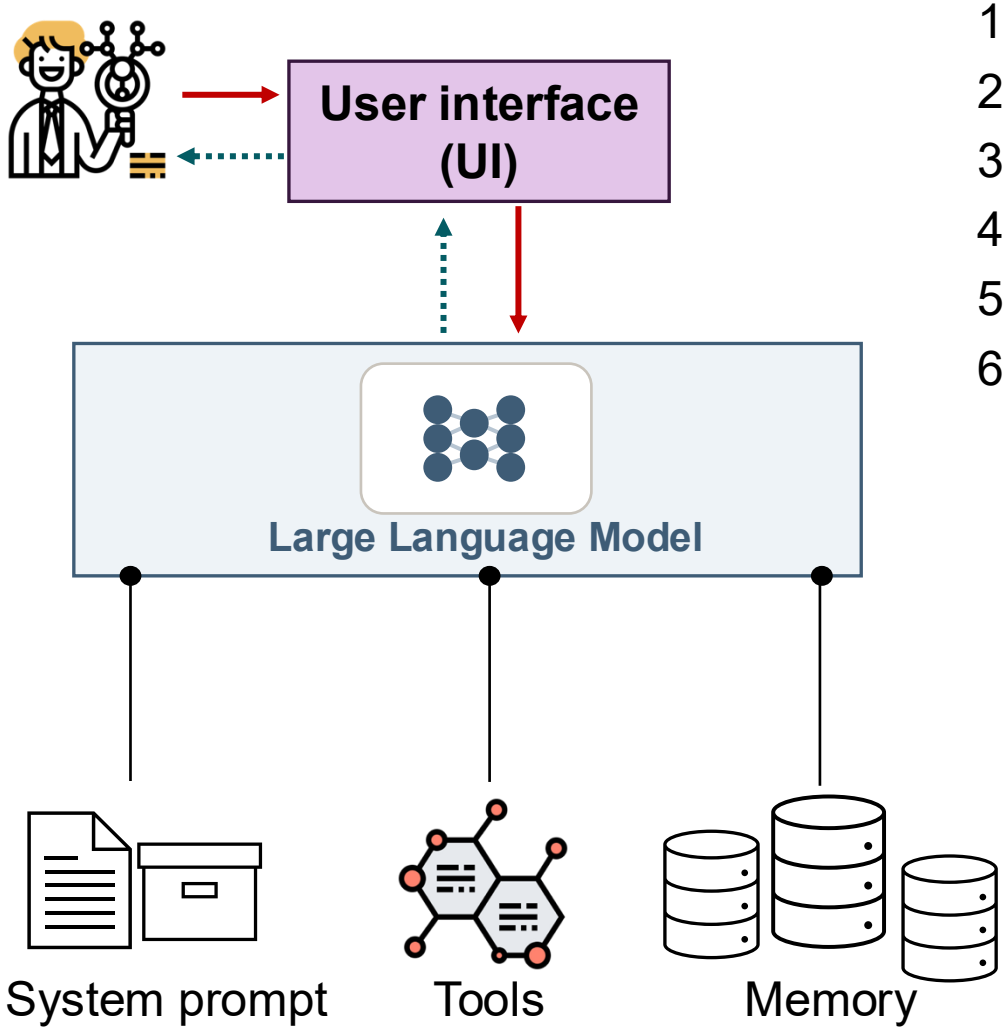
Dinh Long (Chris) Huynh, PhD Student

Department of Pharmaceutical Biosciences, Uppsala University

dinh-long.huynh@uu.se

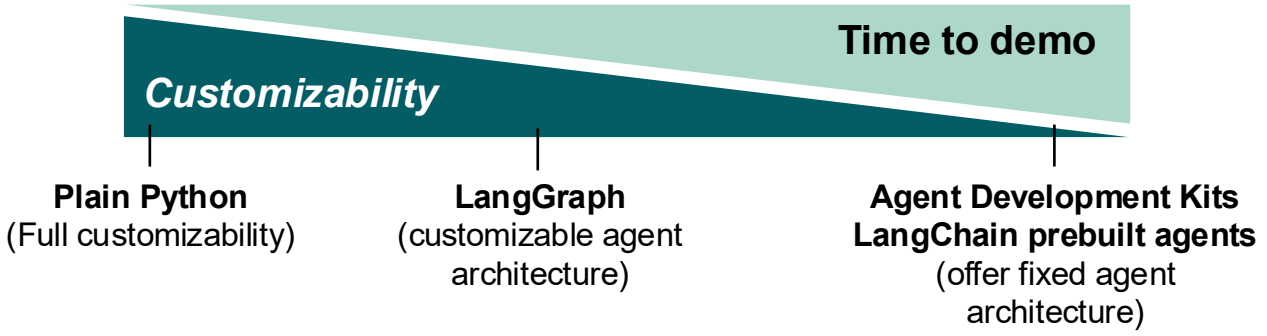
Scan to get the workshop code link

What is Agentic AI system?

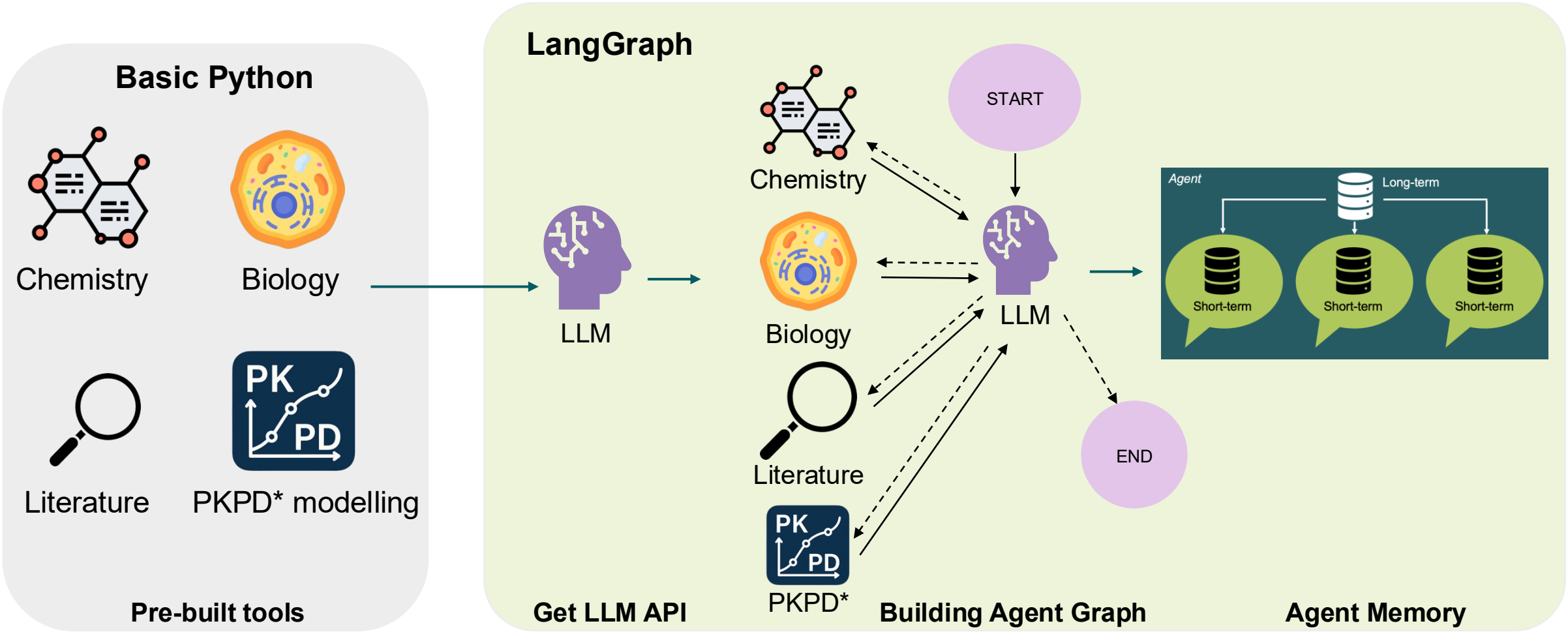


1. Autonomous software system
2. Perceives environment and context
3. Makes decisions
4. Takes actions
5. Achieve specific goals
6. Minimal human intervention

Frameworks for building agents



What are we going to do today?



*PKPD: Pharmacodynamic - Pharmacokinetics

Workshop outline



Part 1: Setup & imports

Part 2: Understanding and creating tools

Part 3: Defining the state

Part 4: Building the agent graph

Part 5: Testing the agent

Part 6: Adding short-term memory

Part 7: Exploring prebuilt agents

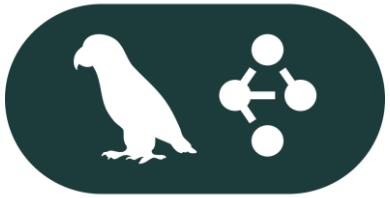
Part 8 (optional): Extension exercises on CodeAct, multi-agent system, long-term memory and human-in-the-loop

About LangChain ecosystem



LangChain

Build chain workflow



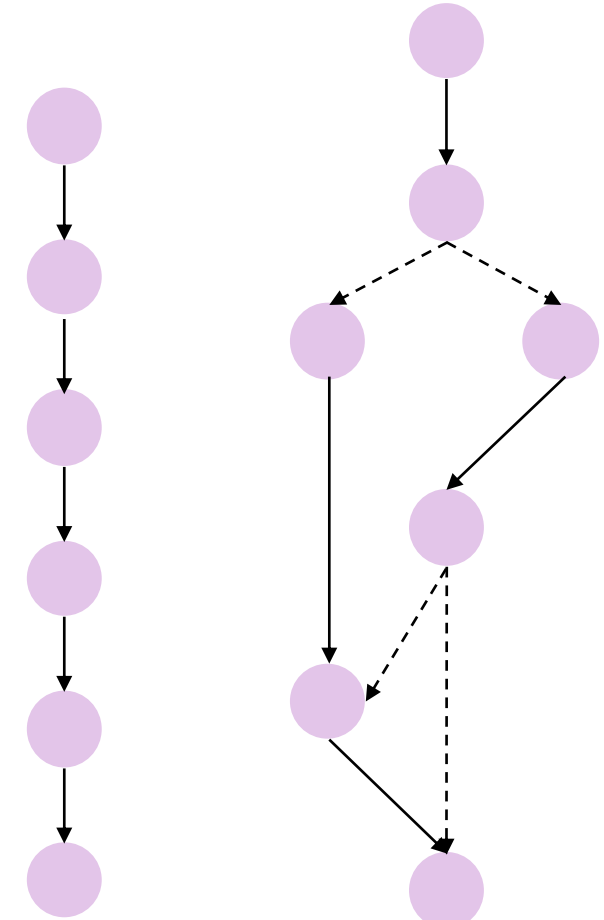
LangGraph

Build graph workflow



LangSmith

Trace running process of all LangChain’s product

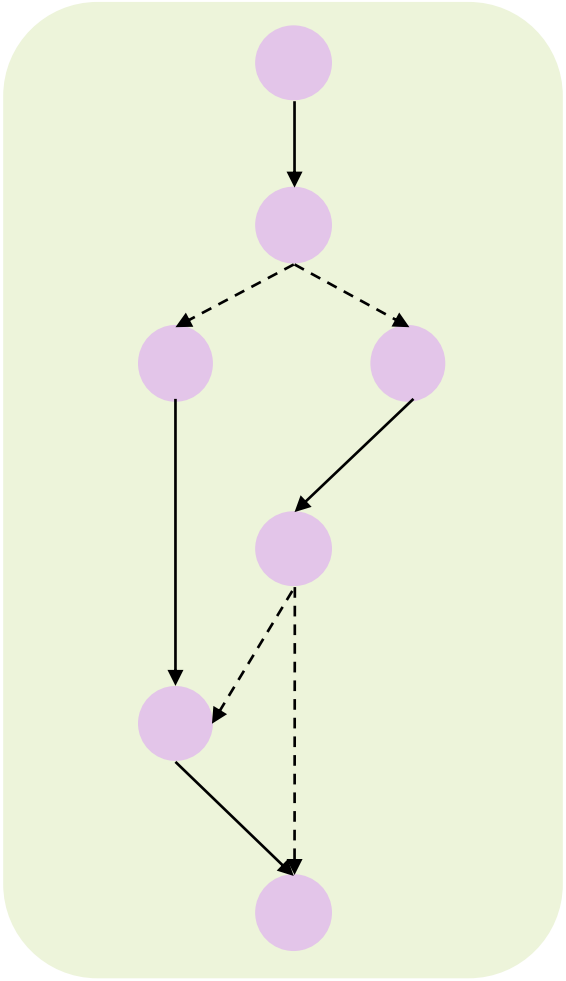


Chain

Graph

LangChain Documentation, 2025

Basic Concepts in LangGraph



Graph

Node A unit of computation: function, tool, pipeline or agent

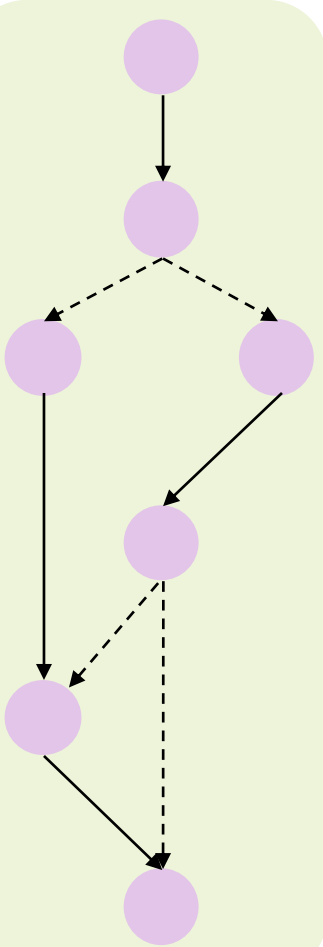
Edge

Static edge → A single compulsory route

Conditional edges ⇢ A flexible route, usually require **decision making** from logical function (if-else), LLM or human

State Shared/Persistent context across the graph. After node invocation, it updates the States.

State in LangGraph



Graph

State Shared/Persistent context across all nodes. After node invocation, it updates the States.

State

messages = [
{'human_message': Hi, How are you today?}
]

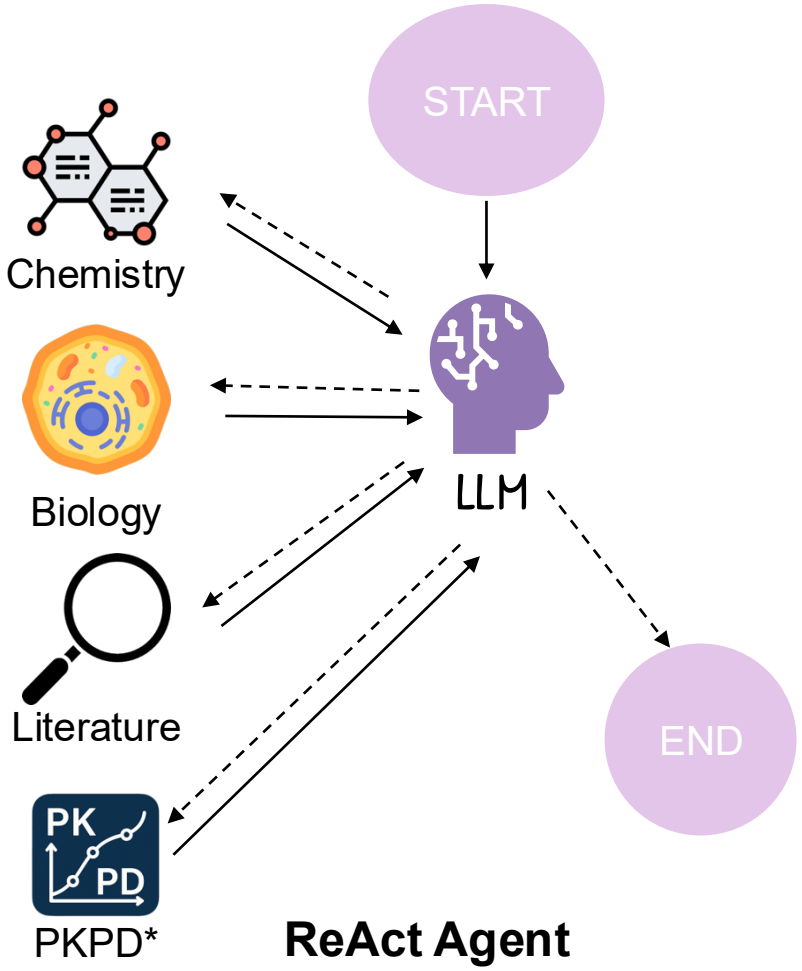
State

messages = [
{'human_message': Hi, How are you today?},
{'ai_message': I am good, what you need me to do?}
]

State

messages = [
{'human_message': 'Hi, How are you today?'},
{'ai_message': 'I am good, what you need me to do?'},
{'human_message': 'Please find me 10 antiviral drugs that can treat COVID-19.'}
]

AI Agent - A simple but flexible graph



ReAct Agent: Reasoning – Acting Agent

- Flexible graph with many conditional edges with a LLM in the center.
- LLM determines which tools to use
- LLM determines when to continue or terminate

Prompts:



" *You are an expert drug discovery researcher. Use your available tools to answer the user's question as accurately as possible. Never fabricate or invent data.* **Which is more lipophilic: Imatinib or Dasatinib?**"

" *You are an expert drug discovery researcher. Use your available tools to answer the user's question as accurately as possible. Never fabricate or invent data.* **What are the common physicochemical characteristics of successful JAK inhibitors?**"

"*You are an expert drug discovery researcher. Use your available tools to answer the user's question as accurately as possible. Never fabricate or invent data.* **Which proteins interact most strongly with EGFR? Search the literature to check whether any of these partners are linked to resistance to EGFR inhibitors such as Erlotinib."**

"*You are an expert drug discovery researcher. Use your available tools to answer the user's question as accurately as possible. Never fabricate or invent data.* **What are the molecular targets and mechanism of action of Imatinib? Describe the biological function and subcellular location of its main target."**

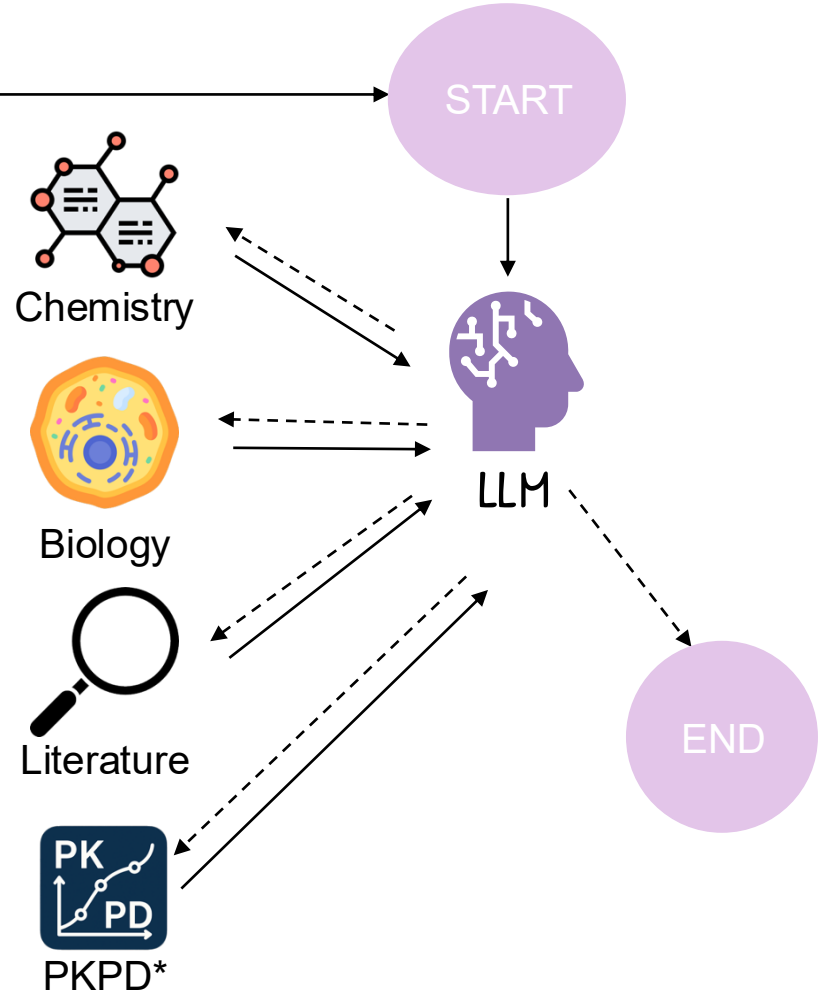
"*You are an expert drug discovery researcher. Use your available tools to answer the user's question as accurately as possible. Never fabricate or invent data.* **Imatinib is given orally at 400 mg once daily. A population PK study (Widmer et al., Br J Clin Pharmacol 2006) reports an apparent oral clearance CL/F = 14.3 L/h and volume V/F = 347 L; assume ka = 0.6 1/h. Simulate two weeks of dosing. Does the steady-state trough stay above 1 mg/L (1000 ng/mL), the threshold associated with major molecular response in CML (Picard et al., Blood 2007)? If not, which once-daily dose, in steps of 100 mg, would reach it?"**

System Prompts:

System prompts: Define agent behaviour

Main Prompts: The request that you want the agent to do

System Prompt	Main Prompts	System Prompt

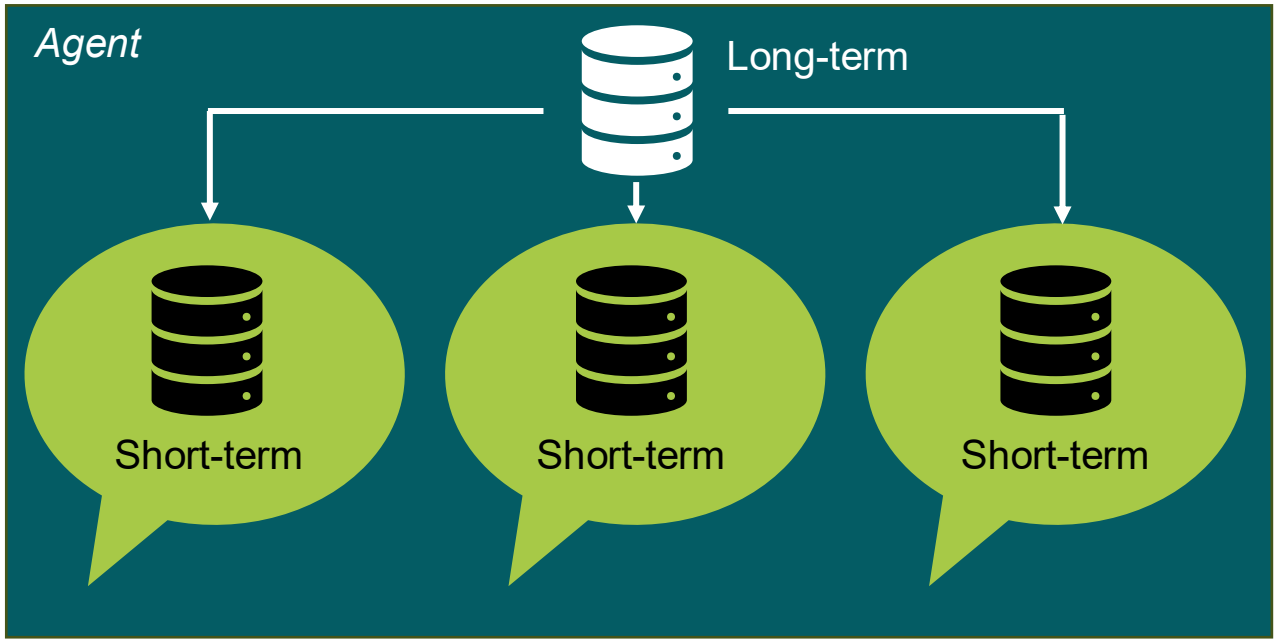


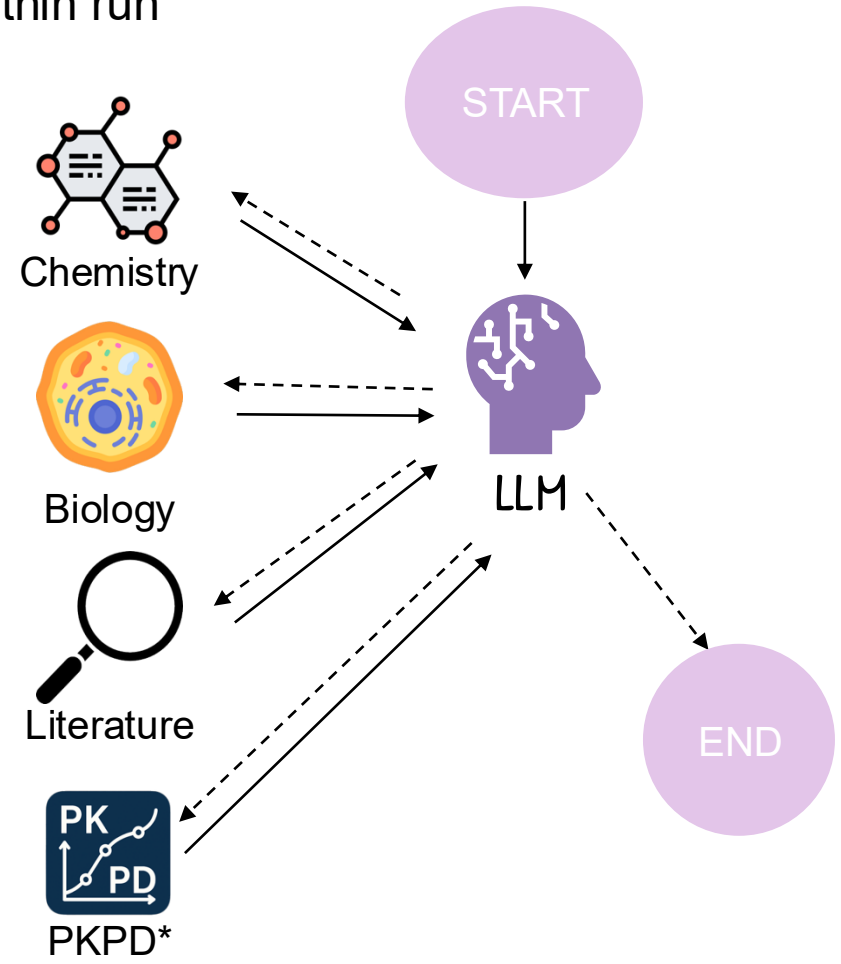
Memory:

- GraphState only maintains context from START to END: Memory within run

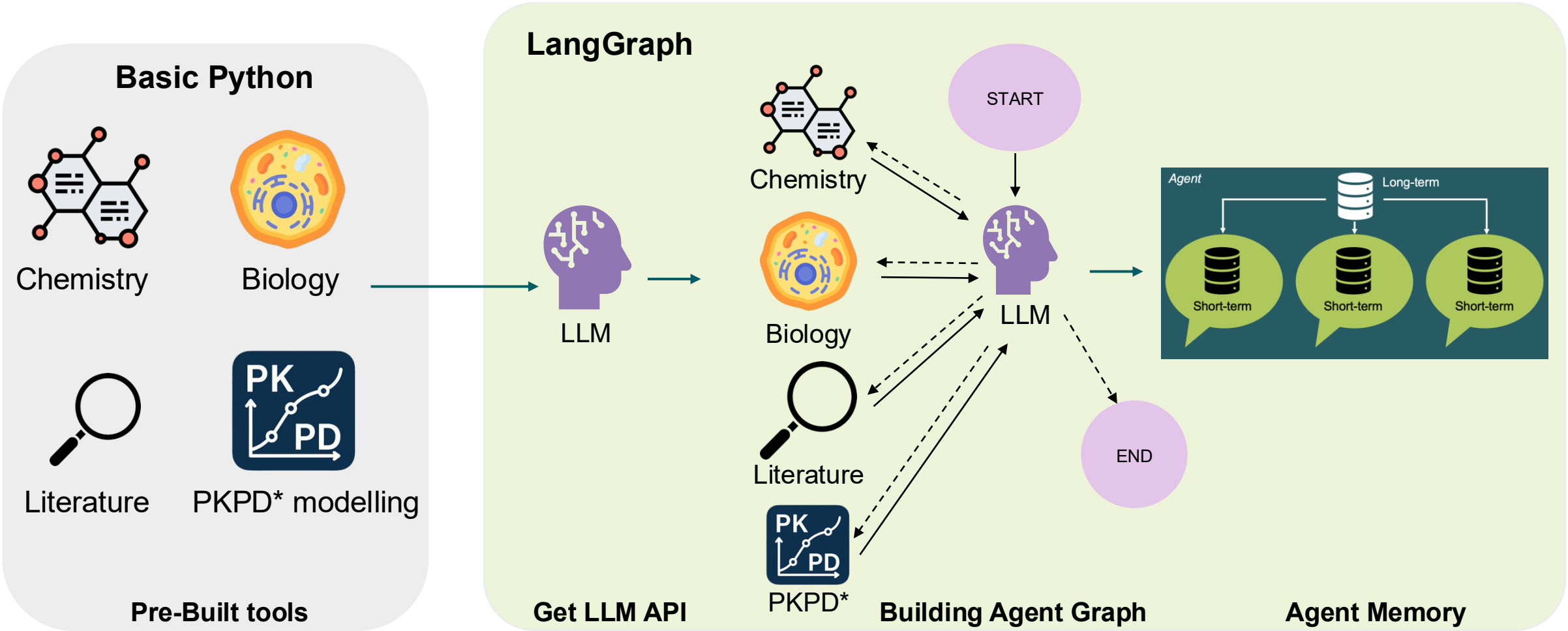
- Short-term memory: Memory within conversation.

- Long-term memory: Memory outside conversation.





What are we going to do today?



*PKPD: Pharmacodynamic - Pharmacokinetics



Thank you for listening!

Feedbacks and Question

Problem with ReAct & Tools



Pre-defined tool is rigid

CCC(CC)COC(=O)C(C)NP(=O)(OCC1C(C(C(O1)(C#N)C2=CC=C3N2N=CN=C3N)O)O)OC4=CC=CC=C4

Remdesivir

CID 121304016

```
@tool
def awesome_tool(smiles: str):
…
```

Fixed tool space

- Tool 1
- Tool 4
- Tool 2
- Tool 3

Scaling: Agent works well with 3-4 tools does not means it works well with 30-40 tools

```
@tool
def tool1():
    ''' Tool description '''
```

```
@tool
def tool2():
    ''' Tool description '''
```

llm.bind_tool(TOOLS)

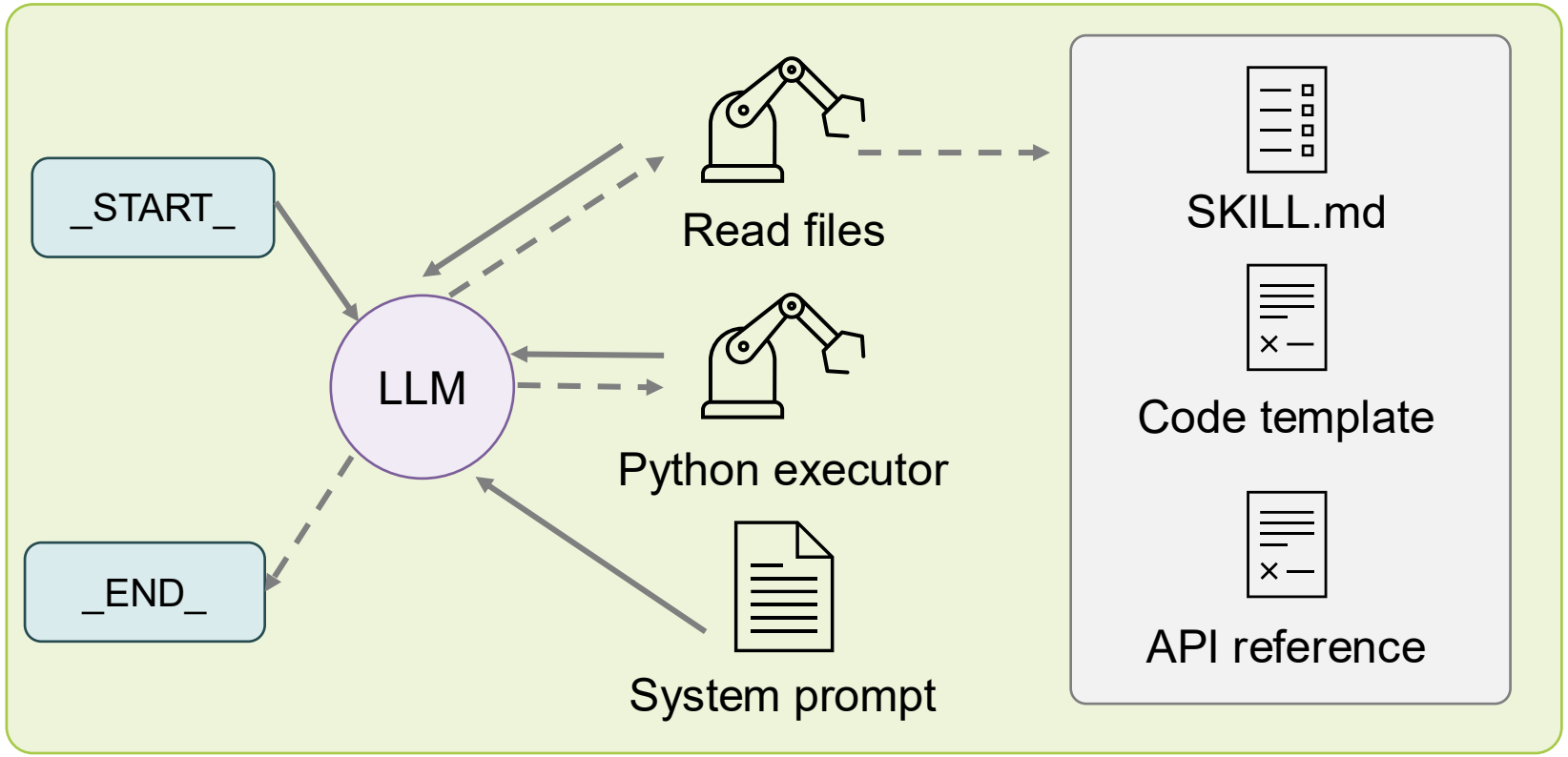
System prompt

Available tools:

tool1: Description

tool2: Description

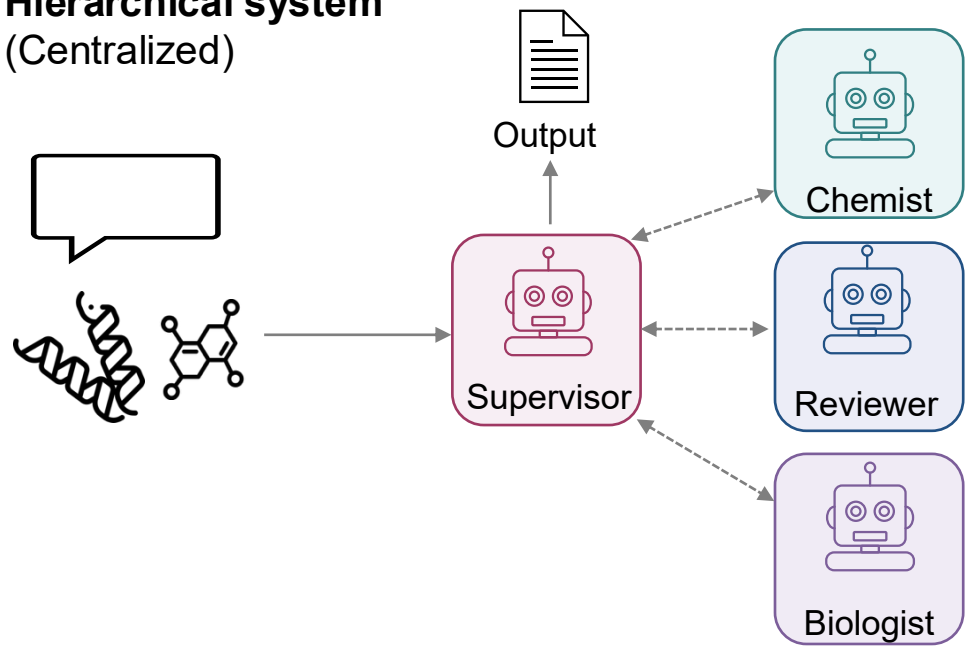
CodeAct Agent



Wang, X. et. al, *Executable code actions elicit better LLM agents* (arXiv:2402.01030). arXiv. https://arxiv.org/abs/2402.01030

Multi-agentic AI systems

Hierarchical system
(Centralized)



Swarm system
(Decentralized)

